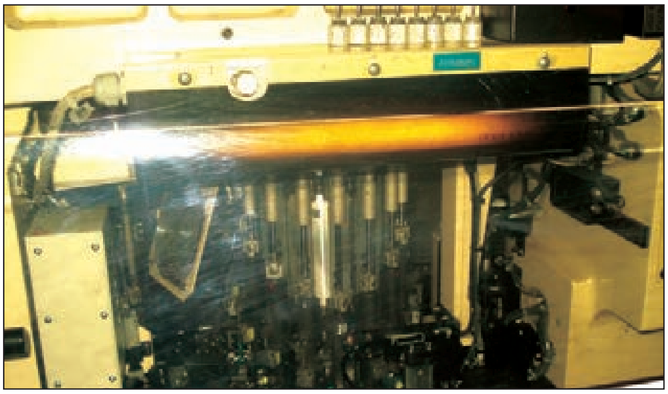
Microsoft + GM
Automobile giant GM is partnering with Microsoft to promote its electric car Volt through multi screen campaign which also taps the new Kinect for Xbox 360

Bing scores ahead of Google
While Google plans to debut its online music service this year, bing scored by showcasing the entertainment features which includes music



Component gun

The component gun can mount as many as eight components in one second.

Then a process of reflow soldering is done, followed by a test and inspection stage. The human element is ever important, on the right is an employee in the process of visual inspection.

If you'll notice, this stage is mostly completed by machinery and is automated. The only human element is involved with the inspection.

At this point we realised the futility of photos to describe what is, essentially a moving work process and switched to video shooting. Please check out www.thinkdigit.com/d/1007_gigabyte for some more videos highlighting important points of the manufacturing process.

DIL: An acronym for Dual in-line package, this is a stage where larger components like the expansion slots resistors and capacitors and chokes are fitted on to the motherboard. These components are put on to the motherboard by hand. All these components mount on the motherboard via insertion into precision holes on the PCB, and are soldered from the other side, which is why the name i.e. dual in-line. This process has a lot of manual labour and there are parallel lines of people sitting in long rows plugging in these components on to the motherboard. The level of concentration required is rather amazing, as is their ability to work mechanically. They have boxes filled with various components, and each person will attach up to six components on a single motherboard. The line never stops, so they are working on a motherboard that is slowly moving along an assembly line. At the end of the line when all the components are mounted, there is a process called Wave soldering that is done to fix these loosely inserted components. As the name suggest a wave of liquid solder is washed on to the bottom side of the PCB, coating the metal pins of the inserted parts.

This part of the process is pretty much the same for the graphic cards. All the components including capacitors and even the power connectors are manually inserted and then the cards go through the wave soldering process. When the motherboards emerge, a thorough quality testing is done through visual inspection.

Final Testing: At this stage each component is tested, for any sort of malfunction. There is a set of quality control standards in place, and each motherboard is functionally checked. There are different parts to this testing from visual inspection to individual component testing, where a semi-automatic machine is pressed down, inserting expansion cards into various ports and slots, checking for proper functioning. There is also a burn-in testing that is done to two per cent of the motherboards in each batch. We were told that problems are divided into two parts – process based problems, where a fault develops in the assembling process, and component-based problems, where one of the components fails. In the latter case, the production process might be stopped if a batch of parts used in the motherboards is found to be faulty. During retesting of such a batch, 5 per cent of the boards are tested for burn-in, and this can be escalated to as much as eight per cent.

Packaging and bundling: This is another process that involves a lot of manpower. Flat boxes are seen along a production line where automated machinery shapes the carton. On another line, the motherboards get some final stickers like batch numbers, dates and so on. Once this is done, the motherboards are packed in antistatic bags and the boxes are populated with accesories like the driver CDs, SATA cables and so on. Then a specific number of boxes are packed in a larger cardboard box and this box is weighed and then sealed after the required designations are printed on the sides. There is an inspection done after this to ensure packaging is proper and all the markings are correct. Boards then go into the warehouse or are directly shipped out to various destinations, as the case may be.

Throughout, the workplace was well lit, and extremely clean and airy. Employee comfort is important to Gigabyte, and this is evident from the environment. The human labour on the production line remains a largely rinse and repeat affair, and is considered a labour (read blue collar) job in Taiwan. However, factories such as these employ people by the hundreds, and we counted up to 95 people on a single motherboard line, and the Nan-Ping factory had eleven such product lines. We're told Gigabyte has another factory a mere ten minutes away, and there are two larger factories in mainland China. These four factories are responsible for supplying Gigabyte motherboards around the world. As they say, every motherboard has a story... d

Visual inspection

Research In Motion has sold its 100 millionth BlackBerry phone. Hinting at two more devices in the near future d Hulu to become part of the Sony PS network soon